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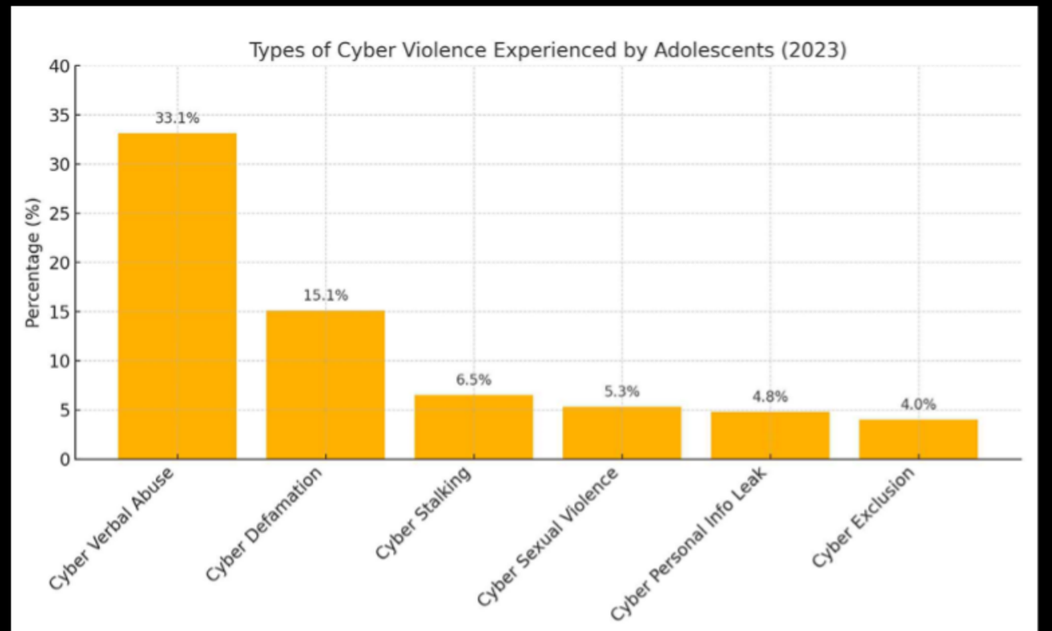
초등학생들의 희망 직업 1위는 4년째 운동선수(9.8%)가 차지하고 있다. 2위는 1년 전보다 한 계단 상승한 교사(6.5%)였다. 3위에는 크리에이터(6.1%), 4위에는 의사(6.0%), 5위에는 경찰관/수사관(4.5%)이 각각 올랐다.

Cyberbullying and Anonymity Protection

* Key Summary

33.1% of teenagers have experienced cyber verbal abuse (NIA 2023)

VTubers as a means of protecting privacy



<그림 2> 2023년 사이버폭력 실태조사 그래프

VTubers as a way to express unique personal identities

*** Key Summary**

Expansion of VTuber Culture After the Pandemic

Enables content creation without revealing one's face

Utilized on metaverse platforms (e.g., ZEP, ifland)





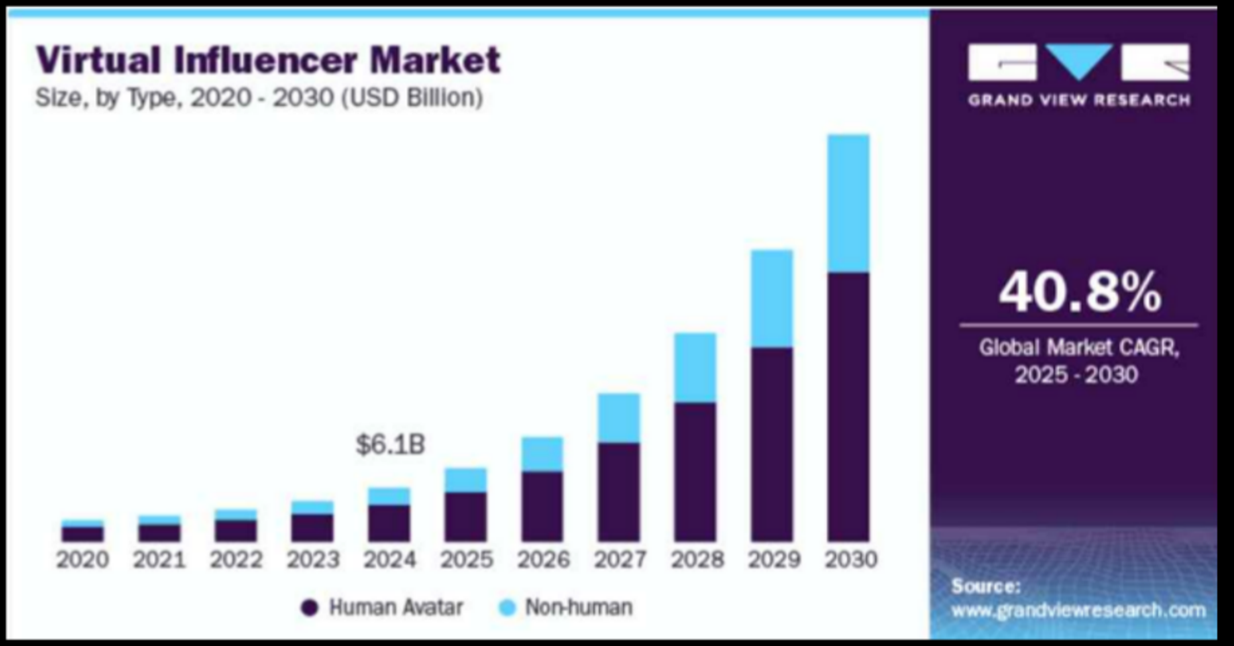


Explosive Growth of the Virtual Market

*** Key Summary**

2024: $6.06 billion
→ 2030: Annual growth rate of 40.8%

SNS Insider: Expected to reach $154.6 billion by 2032



Economic Barriers to Entry for Virtual Model Creation

*** Key Summary**

LIVE2D commissions: Minimum of 100,000 KRW to over 1,000,000 KRW

Equipment + software can cost several million KRW

항목	설명	예상비용
 2D 모델 제작	Live2D돌 활용한 2D 아바타 제작	$100~$200
 3D 모델 제작	Vroid Studio 등으로 제작한 3D 아바타	$1000~ $5000
 하드웨어	PC, 웹캠 , 마이크 등, 장비	$500~$2000
 소프트웨어	Live2D ,Vtube Studio 등의 라이센스	$20~$60 /월

Virtual Model Creation



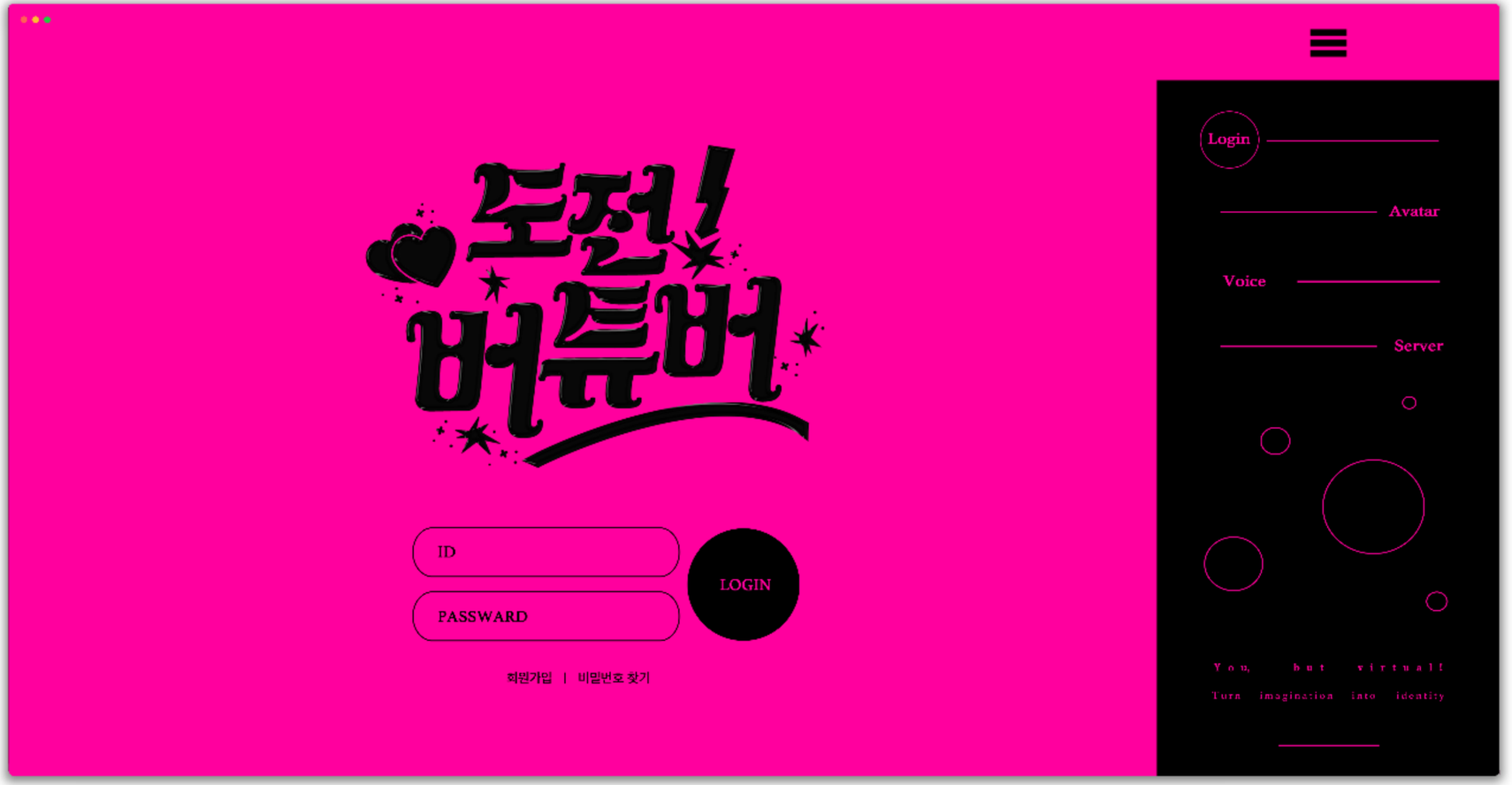
버추얼 아바타 캐릭터를 만들어줘. 걍 보편적인 예쁘고 특이한 느낌으로

2D to 2.5D Conversion

There is a change!



Final program UI example



Final program UI example



Expected Transmission Screen



2D to 2.5D Conversion

There is a change!



Voice Modulation





03. Novelties of the design

Category	Virtual Model Creation	Expression-Based Model Conversion	Text Input-Based Model Creation	Voice Modification	Voice Blending	UX/UI Convenience	Expandability	Cost & Accessibility	System Integration
System (e.g., VSeeFace, FaceRig	Only pre-made, rigged characters can be used	Only basic expressions supported, no auto changes	X	Requires external voice changer	Requires external system	Complex setup, needs additional software	Supports adding characters and modifying poses	Some software requires payment	Supports facial tracking and character export (voice modification needs external program)
iPhone Memoji/ ARKit Emoji	Fixed models only, no creation capability	Expression tracking available, but model conversion not possible	X	X	X	Simple setup, lacks expandability	Limited to Apple devices, lacks external expansion	Device-dependent, limited functionality	Supports tracking only, no export function
Auto rig pro	X	Available, semi-automated	X	X	X	Complex setup, Blender usage is complicated	Can rig other characters	Paid	Only semi-automated rigging supported
Meshy ai	Available (image to 3D model)	Available Not face rigging	Available	X	X	Easy prompt input and UI configuration	Expandable with various formats	Paid	Only generates 3D models from images

Category	Virtual Model Creation	Expression-Based Model Conversion	Text Input-Based Model Creation	Voice Modification	Voice Blending	UX/UI Convenience	Expandability	Cost & Accessibility	System Integration
Applio	X	X	X	Supported, but no real-time output	Voice blending supported	Limited convenience	Real-time output not supported	Free, but only offers blending feature	Voice only, no video
*Voice Changer (e.g., w-okada, used standalone)	X	X	X	Voice modification only	Voice blending supported	Limited convenience	No AR support	Free, but only offers voice modification	Voice only, no video
Live2D	2D virtual model creation specialized	Expression-based animation supported	X	No built-in voice features	No voice features	Professional UI with dedicated software	Extensive plugin support, OBS integration	Paid software (Cubism PRO required)	2D animation + streaming Integration
제안 시스템 (도전 버튜버!)	**Based on LLM (ChatGPT/DALL·E), auto image generation with AR**	**Real-time BlendShapes → Emoji conversion**	**Text input → LLM → Auto image generation**	**Real-time AI-based voice transformation**	**Voice blending supported**	**Intuitive UI for settings**	**Custom prompts, supports various virtual model formats, 2D masking available**	**Free open-source model use, API planned**	**Face tracking + image generation + voice modulation + export**

VTuber System Performance Analysis

📌 **Cost Calculation Basis:**
- Traditional System: Mid-tier Commission ($750) + Live2D Cubism PRO ($95/year) + Rigging ($500) + PC ($1,000) + iPhone ($700)
- Our System: Runpod GPU ($100) + PC ($1,000) + iPhone ($700)
- Required Software: Traditional 3 (Live2D Cubism, VTube Studio, OBS) → Our 1 (Integrated System)
- All prices based on 2025 market research (Live2D official, Fiverr, VGen, Live3D, etc.)

01. Key Performance Indicators (KPIs)

COST REDUCTION

41.5%

$3,045 → $1,800

$1,245 saved annually

TIME REDUCTION

95.8%

144hrs → 6hrs

138 hours saved

NO HARDWARE

92.3%

$1,345 → $100

With existing PC/iPhone

SOFTWARE REQUIRED

67%

3 → 1

Reduction

LEARNING TIME

87.5%

60hrs → 7.5hrs

Reduction

ANNUAL ROI

1,245%

Investment $100

1-year savings $1,245

02. Scenario-based Cost Comparison (2025 Market Prices)

Scenario	Traditional Solution	Our System	Savings	Reduction
With Existing Hardware (PC + iPhone owned)	$1,345 (Commission $750 + Rigging $500 + Live2D $95)	**$100 (Runpod GPU credits)**	**$1,245**	**92.6%**
PC Only (Need to buy iPhone)	$2,045 (Above + iPhone $700)	**$800 (Runpod $100 + iPhone $700)**	**$1,245**	**60.9%**
Complete Fresh Start (Buy all hardware)	$3,045 (Above + PC $1,000)	**$1,800 (Runpod $100 + PC $1,000 + iPhone $700)**	**$1,245**	**40.9%**

💡 **Real Use Case:** Most users already own a PC, and many have a Face ID-enabled iPhone, resulting in actual savings of 90%+ in these cases.

03. Detailed Feature Comparison

Comparison Item	Traditional Solution	Live2D + Commission	Our System
Initial Cost (Excl. Hardware)	$800-$2,000 (Avg. $1,345)	$500-$1,500 (Avg. $1,000)	**$50-$200 (Avg. $100)**
Monthly Maintenance	$0-$20 (Updates/features)	$8/month (Live2D PRO)	**$0-$10/month (GPU usage)**
Creation Time	48-96 hours (2-4 days)	120-240 hours (5-10 days)	**6 hours**
Technical Difficulty (0-100)	70 points (Intermediate+)	85 points (Expert required)	**25 points (General users)**
Required Hardware	PC ($800-$1,500) Webcam ($50-$150)	PC ($1,000-$2,000) Webcam or iPhone	**PC ($1,000+) iPhone ($700+, Face ID)**
Customization Freedom (0-100)	40 points (Preset limited)	80 points (High but complex)	**95 points (Very high & simple)**
Real-time Processing Delay	50-100ms (Avg. 75ms)	80-150ms (Avg. 115ms)	**<100ms (Avg. 85ms)**

04. AI-based Automation Performance

Work Stage	Traditional Method (Time)	Traditional Cost	Our Method (Time)	Our Cost	Time Saved	Cost Saved
Character Design	48-120 hours	$500-$1,000	**0.5 hours**	**$5-$15**	**99.6%**	**98.5%**
Rigging Work	16-40 hours	$300-$800	**0.1 hours**	**$2-$5**	**99.8%**	**99.3%**
Expression Setup	8-20 hours	$200-$400	**Auto-mapped**	**$0**	**100%**	**100%**
Voice Model Training	4-8 hours	$100-$200	**0.3 hours**	**$3-$8**	**96.3%**	**96%**
Integration & Testing	16-24 hours	$200-$400	**5 hours**	**$0**	**79.2%**	**100%**
Total Process	**92-212 hours**	**$1,300-$2,800**	**6 hours**	**$10-$35**	**97.2%**	**98.8%**

05. Real-time Performance Metrics

Performance Metric	Target Value	Actual Achievement	Industry Average	Assessment
Face Tracking FPS	60fps	**30fps**	30-45fps	Standard Level
CPU Usage	<40%	**20-28%**	45-60%	**-50% More Efficient**
Memory Usage	<4GB	**2.2-2.8GB**	4.5-6GB	**-53% More Efficient**
GPU Usage (Real-time)	<50%	**25-35%**	55-70%	**-43% More Efficient**
Total Pipeline Latency	<150ms	**140-180ms**	180-250ms	**-16% Improved**

💡 **Our System's Strength:** While face tracking FPS is at standard level, **CPU/Memory/GPU usage is 40-50%+ lower than industry average,** allowing smooth simultaneous game streaming or other tasks.

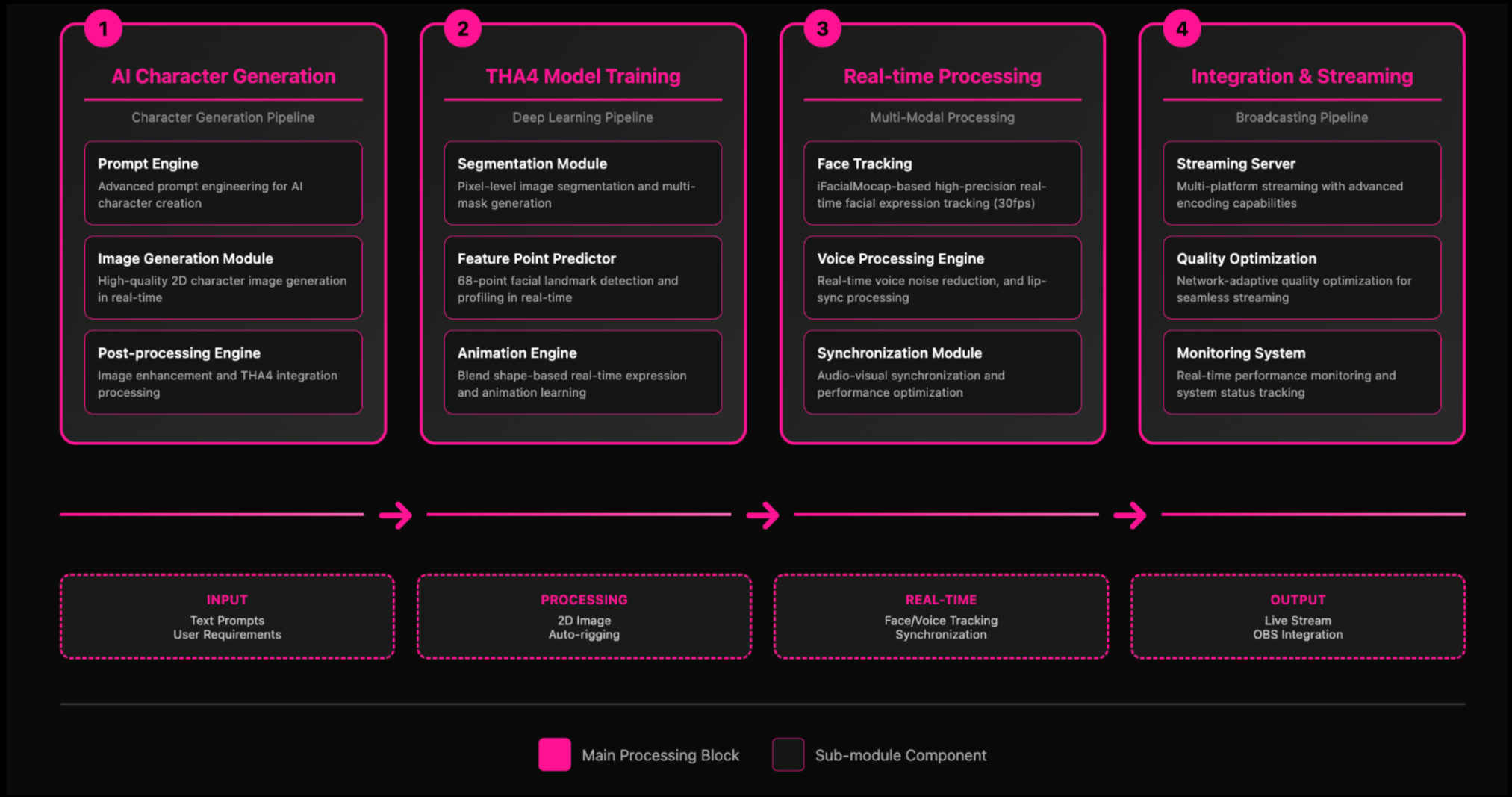
07. Return on Investment (ROI) Analysis

Period	Traditional System Cost	Our System Cost	Savings	Reduction
Initial Investment (With Existing Hardware)	$1,345	**$100**	**$1,245**	**92.6%**
Year 1 (Complete Fresh Start)	$3,165 (Initial $3,045 + Maint. $120)	**$1,920 (Initial $1,800 + Maint. $120)**	**$1,245**	**39.3%**
Year 3 (Cumulative)	$3,405 ($3,045 + $120×3)	**$2,160 ($1,800 + $120×3)**	**$1,245**	**36.6%**
Year 5 (Cumulative)	$3,645 ($3,045 + $120×5)	**$2,400 ($1,800 + $120×5)**	**$1,245**	**34.2%**
ROI (With Existing Hardware)	-	**Investment: $100**	**Savings: $1,245**	**1,245%**

📊 **Sources:**
- Live2D Cubism PRO Pricing: Official Website (¥14,280/year ≈ $95)
- VTuber Commission Pricing: Fiverr, VGen, Live3D, VTuber Market averages
- PC Pricing: PCPartPicker, MSI, Market Research
- iPhone Pricing: Apple Official Website (Face ID-enabled models)

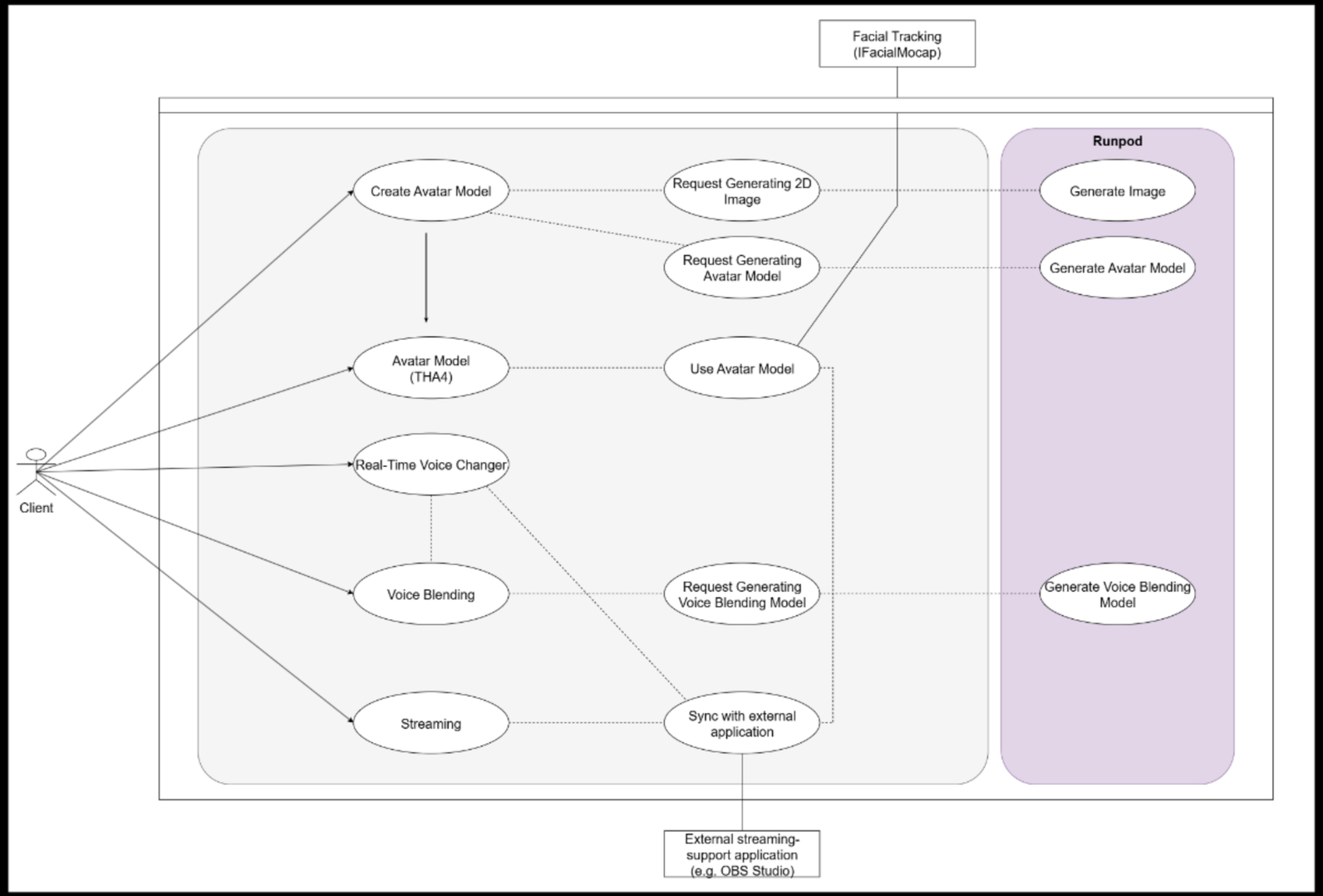
System Block Diagram

There is a change!



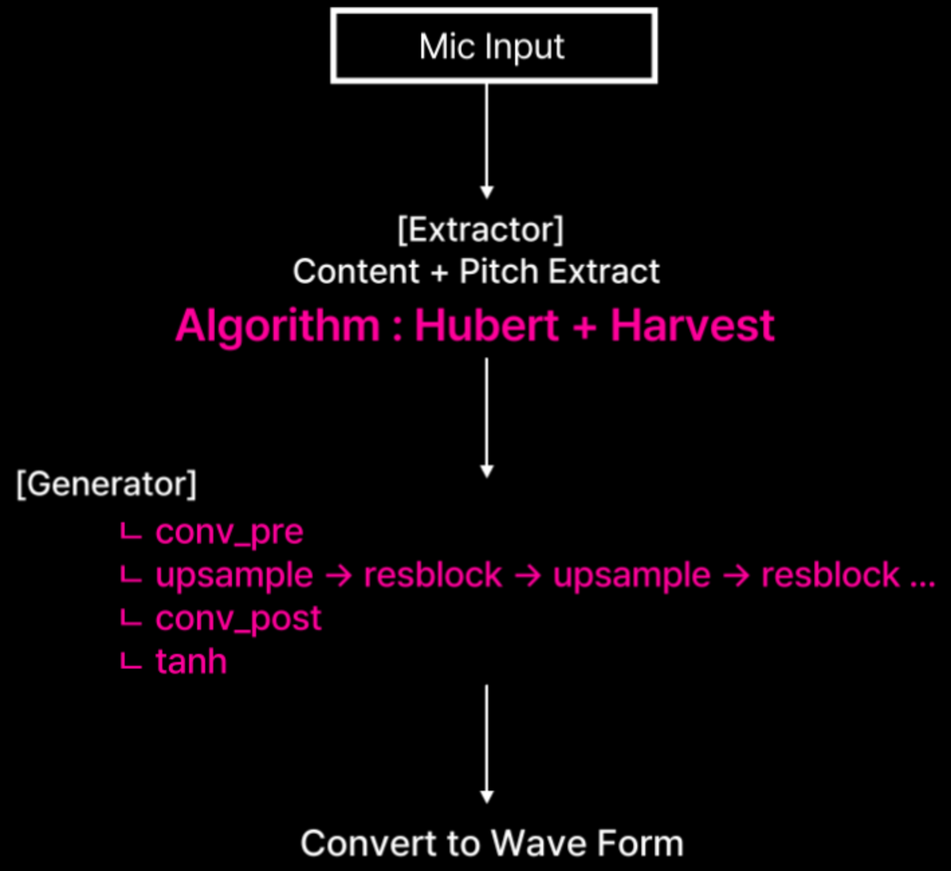
System Block Diagram

There is a change!



Virtual Voice

Real Time Voice Changer



Voice Blending

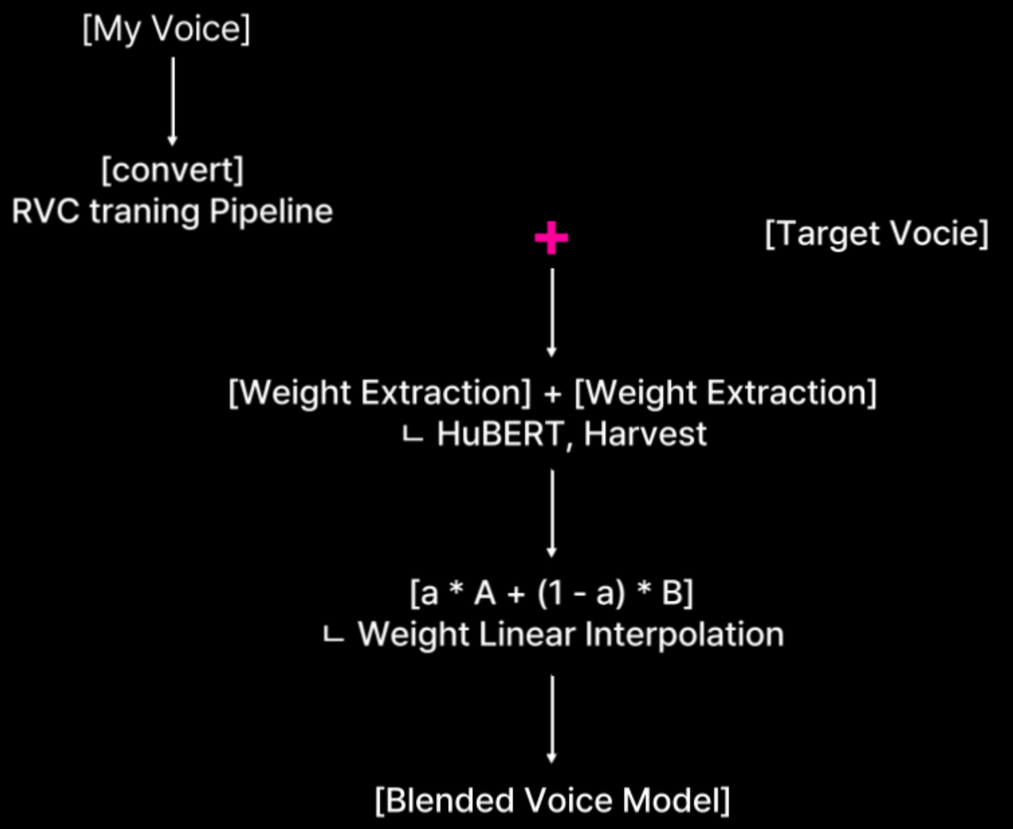


Image Generation

Prompts

1girl, hatsune miku, worst quality, ms paint (medium).



ILLUSTRIOUS

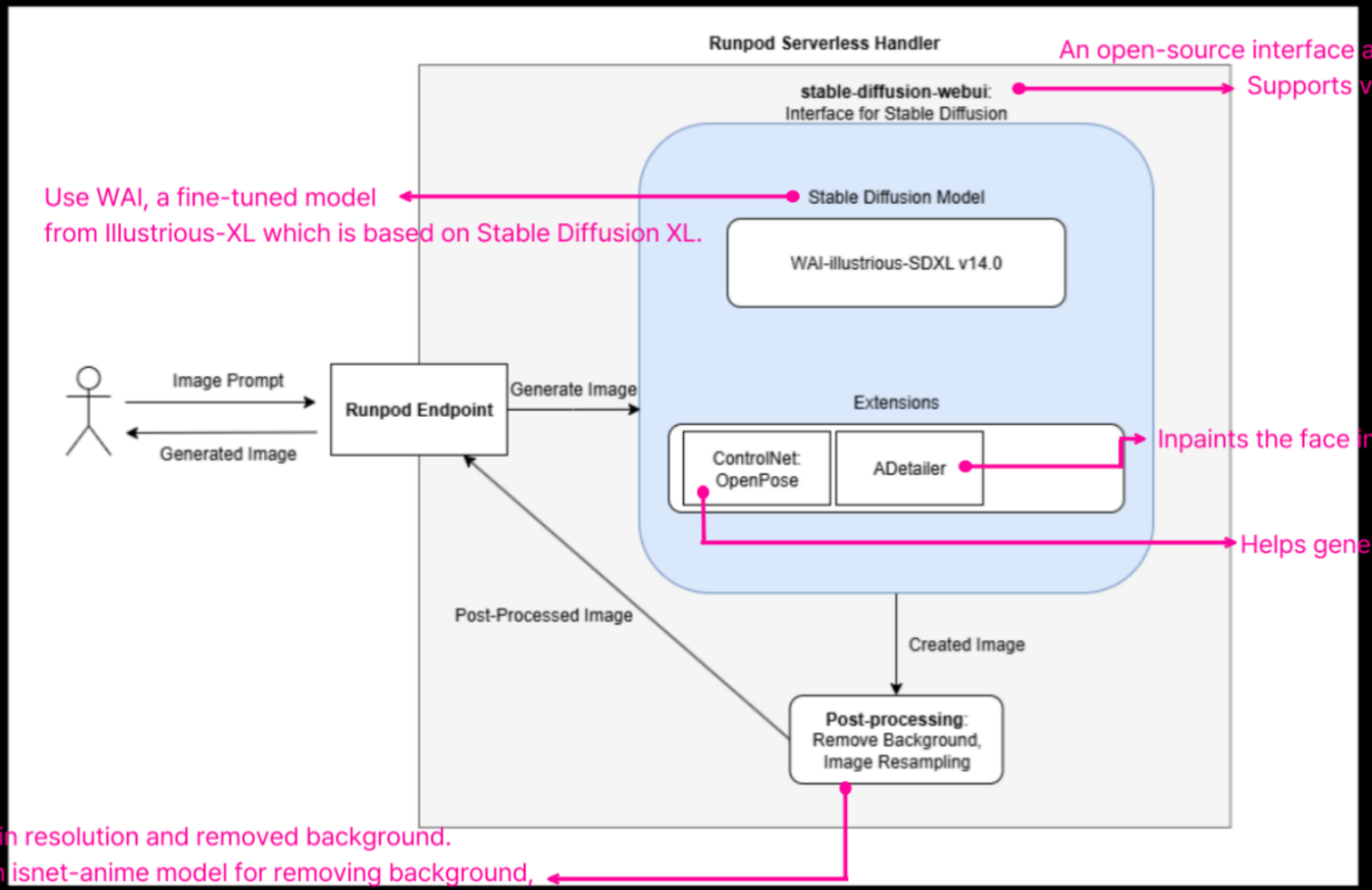
1girl, happy, smile, crying, 2koma.



Fine Tuning (LoRA)

Image Generation

There is a change!

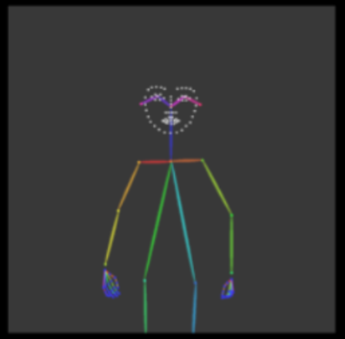


An open-source interface application for Stable Diffusion.
Supports various models and extensions.

Use WAI, a fine-tuned model
from Illustrious-XL which is based on Stable Diffusion XL.

Inpaints the face in the generated image for better quality.

Helps generate images of desired poses.

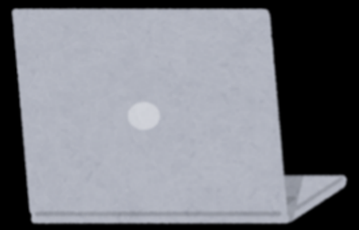


To generate a model with THA4,
an input image should have certain resolution and removed background.
We use rembg Python library with isnet-anime model for removing background,
and Pillow for image resampling.

Image Generation

There is a change!

Write and send image prompt



User

Request

Response (202)

Retrieve image (later)

Firebase-based Backend
Validate and forward request



Firebase

Request

Response

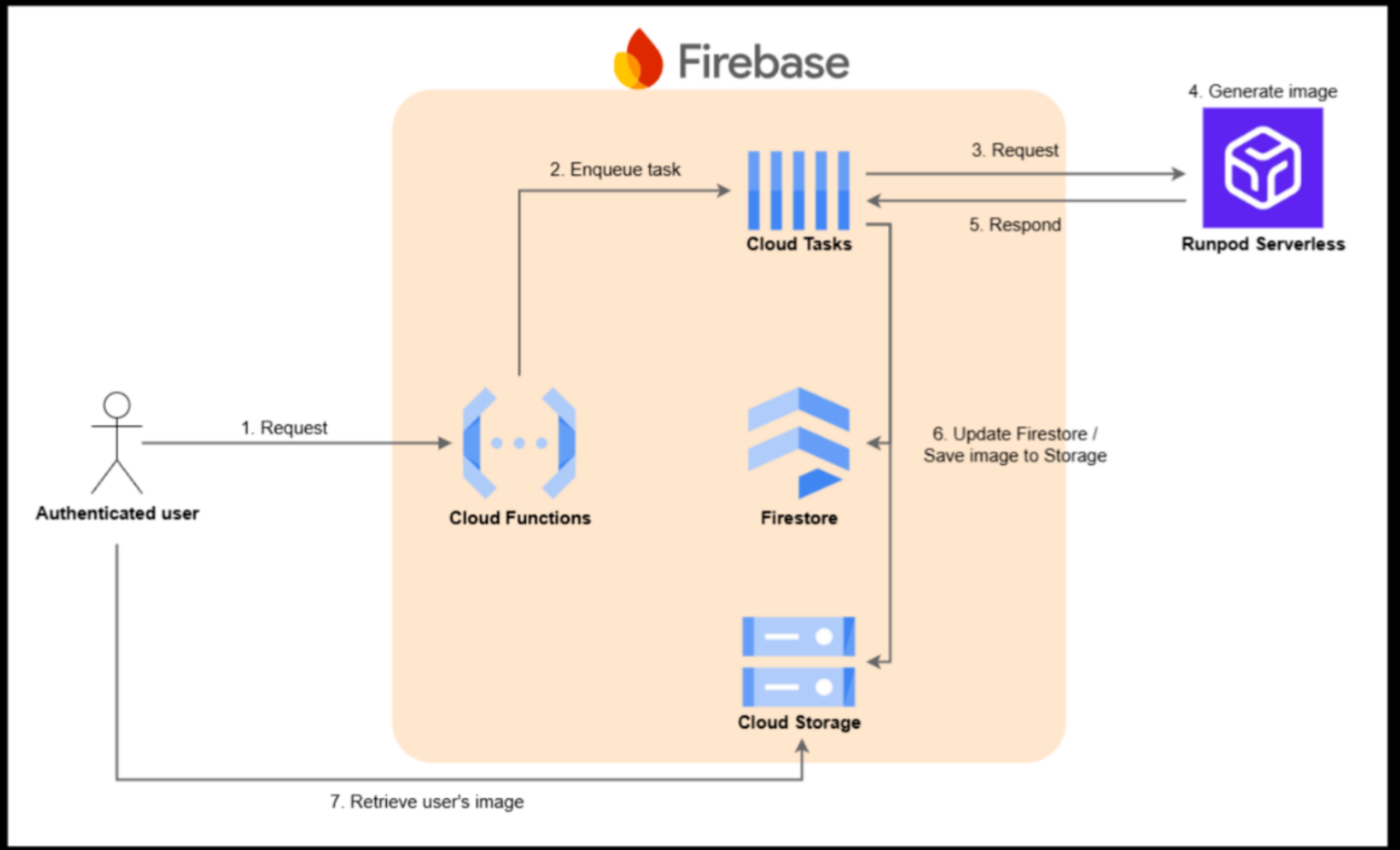
Deploy AI image generator to Runpod
Generate requested image



Runpod

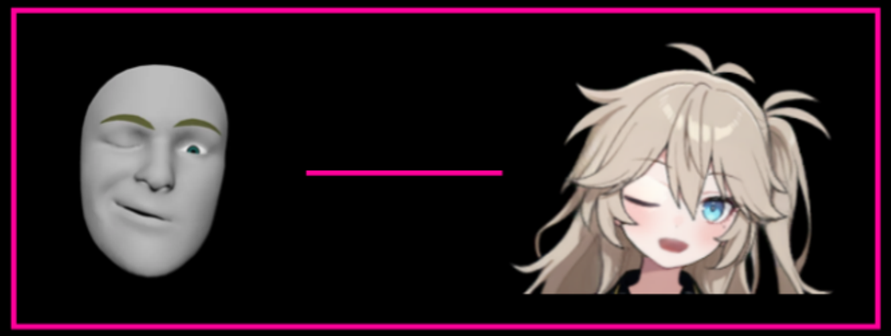
Image Generation

There is a change!



Modeling

There is a change!





IFACIALMOCAP

Face motion

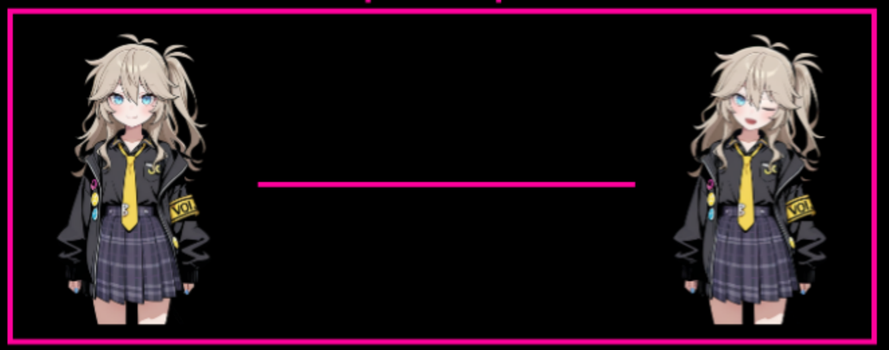
pixiv

Your creative journey awaits

Taking head motion



OBS studio

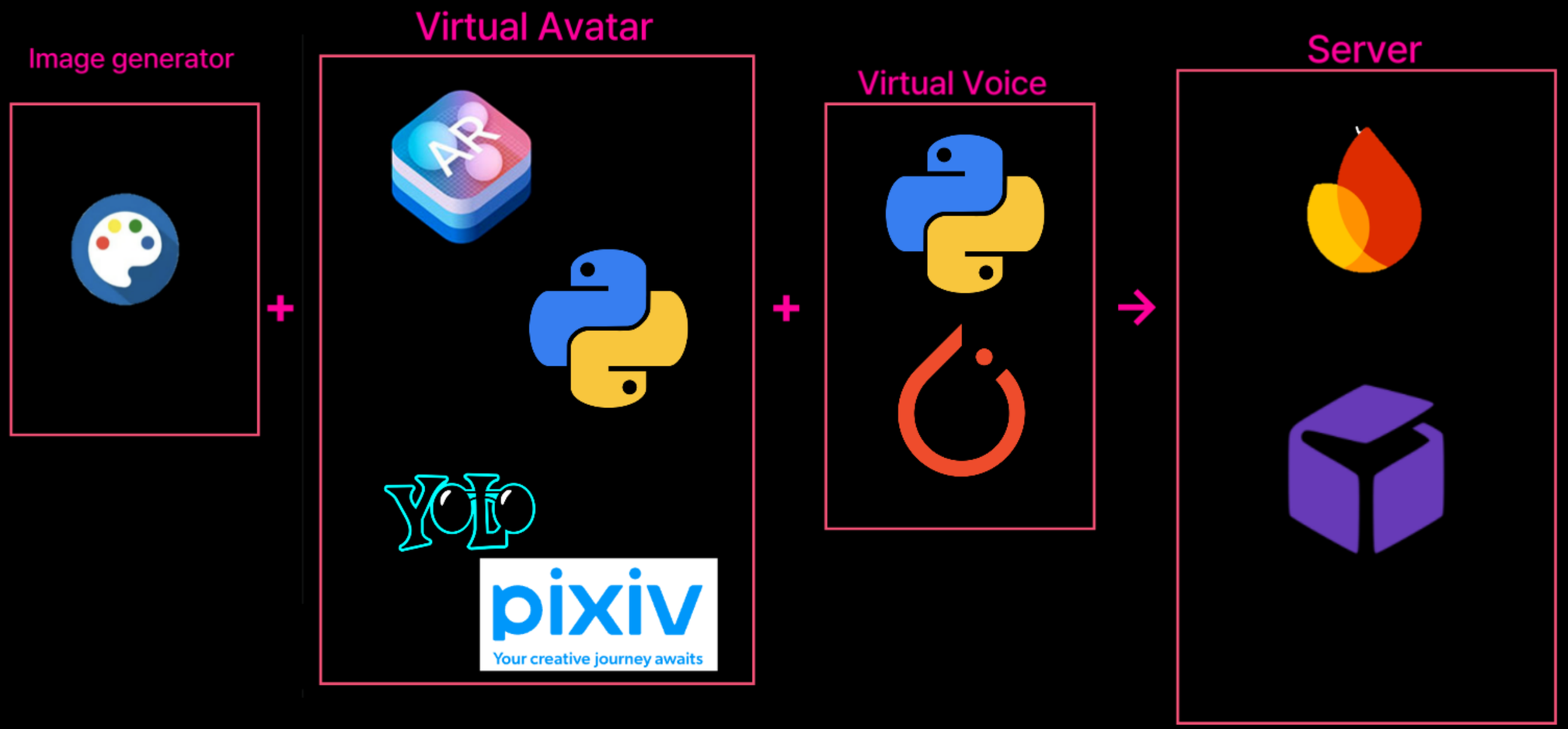


Image

Capture

Virtual Avatar

There is a change!



Task Allocation for each team member

01 방문혁

AR integration, automated image rigging,
iPhone compatibility

02 한승호

Image rigging model tuning,
Server implementation

03 조현석

Voice modulation,
Server implementation

04 김리현

UI development,
LLM model training

Work Schedule

Task Description	May	Jun	Jul	Aug	Sep	Oct	Nov
Image and model generation based on user request							
Real-time voice changer and voice blending implementation							
Development of auto-rigging system							
Backend system construction							
Development of user application							

Target Performance Metrics

Functions	Detailed Items
User APP	- Animates model based on face tracking and displays the result in the screen
Real-Time Voice Changer	- Performs real-time voice modification using voice feature mapping, with noise filtering
Voice Blending	- Blends the user's voice with a target voice based on tone, without altering pitch
Creating VTuber Model	- Generates a character image from LLM-processed text prompts, and creates a model from the generated image
Auto-Rigging System	- Automatically performs facial rigging by segmenting facial features and attaching rigging points to each feature
Server	- Loads AI models and the database, then runs inference on the models based on user's inputs

Performance Evaluation Method

Functions	Detailed Items
Real-Time Voice Changer	- Latency: Voice delay from input to output. Should be under 100ms for smooth real-time use. - PESQ: Checks how natural the voice sounds after conversion. Over 3.5 means good quality. - STOI: Measures how clearly the voice can be understood. Over 0.75 is considered clear.
Voice Blending	- Output Stability: We check if voice quality stays stable with different input voices. If scores don’t change much, the model is considered stable.
VTuber Image Generation	- CLIP Score: Measures how well the generated image matches the user’s text prompt (like eye color, hair style). Over 0.30 is a good match.
Auto Rigging	- We compare our face landmark model with other public models to see which works better for animation.